A q-Deformation of Number Theory

2024

Abstract

In this paper we present a q-deformation in number theory. This q-deformation, in order to be consistent, will be derived from the most natural factorization of q-numbers. In this paper we demonstrate the power of this q-deformed number theory, by proving some known identities, as well as some possible new ones (Result 3.2.2, Result 3.2.4).

1 Introduction

We present the elementary building blocks of a very naturally arising q-deformation of number theory, through the study of q-numbers and their properties. The fundamental idea is to study number theory from 'inside the bracket', meaning that for a quantum number $[n]$ we generally start with known properties of n inside the bracket. This allows for the preservation of some structures in number theory. The conventional q-number is defined as

$$[n]_q = 1 + q + \ldots + q^{n-1} = \frac{1-q^n}{1-q}$$

Some authors will use an alternative symmetric definition for a q-number, which just like $[n]_q$ arises in the study of the q-Weyl algbera and takes the following form,

$$\frac{q^n - q^{-n}}{q - q^{-1}}$$

This notation $[n]_q$, whilst popular, becomes messy when dealing with many exponents on q which will become obvious when looking at, for example, the q-analog of the fundamental theorem of arithmetic as touched upon in section 2.2. Due to this I want to introduce a more compact notation for q-numbers. For starters, we will let

$$[n;\alpha] = [n]_{q^\alpha} = \frac{1-(q^\alpha)^n}{1-q^\alpha}$$

if the exponent $\alpha = 1$ we will just write $[n]$. The rules for addition and multiplication for q-numbers are as follows

$$[n+m] = [n] + q^n[m]$$

$$[nm] = [n][m;n]$$

2 Factorization of q-Numbers

2.1 Cyclotomic Factorization

Let $\Phi_n(x)$ denote the n-th cyclotomic polynomial defined as

$$\Phi_n(x) = \prod_{\substack{1\leq k\leq n\\ \gcd(n,k)=1}} (x-\zeta_n^k)$$

where $\zeta_n = e^{2\pi/n}$ is the n-th root of unity. One can derive the very natural formula that is

$$[n] = \prod_{\substack{d|n\\ d>1}} \Phi_d(q) \tag{1}$$

This allows us to construct some q-analogs of arithmetic functions in a natural way (see section 3 and 4).

2.2 Analog of the Fundamental Theorem of Arithmetic

The fundamental theorem of arithmetic states that any natural number n is the unique product of prime numbers up to the order of factors. For example,

$$196883 = 47 \cdot 59 \cdot 71$$

to see this more generally, for any numbers $n, k \in \mathbb{N}$, and p a prime number we can write

$$n = \prod_{i=1}^{\omega(n)} p_i^{k_i}$$

where $\omega(n)$ is the prime omega function. The same is true for q-numbers, namely that they can be factorized uniquely into constituent 'primitive' q-numbers. We can see that

$$[n] = \left[\prod_{i=1}^{\omega(n)} p_i^{k_i}\right] = [p_1^{k_1}][p_2^{k_2}; p_1^{k_1}] \dots [p_i^{k_i}; p_1^{k_1} p_2^{k_2} \dots p_{i-1}^{k_{i-1}}]$$

by the multiplication law for q-numbers. Furthermore, since we are dealing with powers on the prime factors, we can split up the expression into more primitive q-numbers as such

$$[n] = \left(\prod_{j=0}^{k_1-1} [p_1; p_1^j]\right)\left(\prod_{j=0}^{k_2-1} [p_2; p_2^j p_1^{k_1}]\right) \dots \left(\prod_{j=0}^{k_i-1} [p_i; p_i^j \prod_{r<i} p_r^{k_r}]\right)$$

This will allow us to write the expression in such a way that we can naturally factor a q-number as a product of prime cyclotomic polynomials. From the expression obtained above we conclude that

$$[n] = \prod_{i=1}^{\omega(n)} \prod_{j=0}^{k_i-1} [p_i; p_i^j \prod_{r<i} p_r^{k_r}]$$

This is the analog for the fundamental theorem of arithmetic for q-numbers. We can see that the uniqueness of the fundamental theorem of arithmetic still holds. q-numbers add a degree of 'fruitfulness' to numbers, as certain properties of polynomials arise, allowing for potential study of polynomial rings and matrices, however this is not something we will touch upon in this paper. One can easily see that $\Phi_p(q) = [p]$ and subsequently further factorization into cyclotomic polynomials is possible,

$$[n] = \prod_{i=1}^{\omega(n)} \prod_{j=0}^{k_i-1} \Phi_{p_i}\left(q^{p_i^j \prod_{r<i} p_r^{k_r}}\right)$$

Note: Just for amusement,

$$\prod_{i=1}^{\omega(n)} \prod_{j=0}^{k_i-1} \Phi_{p_i}\left(x^{p_i^j \prod_{r<i} p_r^{k_r}}\right) = \prod_{\substack{d|n \\ d>1}} \Phi_d(x)$$

3 q-Deformed Arithmetic Functions

This section is where it seems to get interesting...

q-Von Mangoldt Function

Derivation

Now, we may introduce the so called 'q-Von Mangoldt function' which exhibits many interesting properties worth discussing, as well as a very natural derivation from the q-number's cyclotomic factorization (2.1). We will denote the q-Von Mangoldt function as $\Lambda_n(q)$. Classically, the Von Mangoldt function is defined as

$$\Lambda(n) = \begin{cases} \ln p & \text{if } \ n = p^k \quad \text{for some prime } p \text{ and } \ k \geq 1 \\ 0 & \text{otherwise} \end{cases}$$

A very important identity used throughout number theory is,

$$\ln n = \sum_{d|n} \Lambda(d)$$

If we turn now to q-numbers, we can define a naturally arising quantum number theoretic analog of the Von Mongoldt function, which we will call the q-Von Mongoldt function and denote it as $\Lambda_n(q)$. Recall (1), namely that

$$[n] = \prod_{\substack{d|n \\ d>1}} \Phi_d(q)$$

taking the natural logarithm of both sides we can see that

$$\ln{[n]} = \sum_{\substack{d|n \\ d>1}} \ln \Phi_n(q)$$

comparing this to the identity of the classical Von Mongoldt function, and noticing that $\Lambda(d = 1) = 0$ we get the definition for $\Lambda_n(q)$

Definition 1.

$$\ln{[n]} = \sum_{\substack{d|n \\ d>1}} \Lambda_n(q) \tag{2}$$

One can study this function from many different angles. I will briefly present a few below. We can check that

$$\Lambda_n(1) = \Lambda(n)$$

The q-Von Mangoldt function shares intimate connection with Ramanujan's sums, defined as

$$c_k(n) = \sum_{\substack{1\leq m\leq k \\ \gcd(m,k)=1}} \zeta_k^{mn}$$

which were mentioned in a 1918 paper by Ramanujan [1]

Theorem 3.1. *Let $c_k(n)$ denote Ramanujan's sum and $\Lambda_n(q)$ the q-Von Mangoldt function, then we have*

$$\frac{d}{dq}\Lambda_n(q) = \Lambda_n'(q) = \frac{1}{q^n-1}\sum_{m=1}^{n} q^{m-1}c_n(m)$$

Proof. To prove it we will start with the definition (2) and differentiate both sides to get

$$\frac{d}{dq}\Lambda_n(q) = \Lambda_n'(q) = \sum_{\substack{1\le k\le n\\ \gcd(k,n)=1}} \frac{d}{dq}\ln(q-\zeta_n^k)$$

We can differentiate $\Lambda_n(q)$ quite easily, leaving us with

$$\Lambda_n'(q) = \sum_{\substack{1\le k\le n\\ \gcd(k,n)=1}} \frac{1}{q-\zeta_n^k}$$

using the property of geometric sums

$$= q^{-1}\sum_{\substack{1\le k\le n\\ \gcd(k,n)=1}} \frac{q\zeta_n^{-k}}{q\zeta_n^{-k}-1} = \frac{q^{-1}}{q^n-1}\sum_{\substack{1\le k\le n\\ \gcd(k,n)=1}} q\zeta_n^{-k}\frac{(q\zeta_n^{-k})^n-1}{q\zeta_n^{-k}-1}$$

$$= \frac{q^{-1}}{q^n-1}\sum_{\substack{1\le k\le n\\ \gcd(k,n)=1}}\sum_{m=1}^{n}(q\zeta_n^{-k})^m = \frac{1}{q^n-1}\sum_{m=1}^{n} q^{m-1}\sum_{\substack{1\le k\le n\\ \gcd(k,n)=1}}\zeta_n^{-k}$$

$$= \frac{1}{q^n-1}\sum_{m=1}^{n} q^{m-1}c_n(-m)$$

And by the symmetry of Ramanujan's sums,

$$\Lambda_n'(q) = \frac{1}{q^n-1}\sum_{m=1}^{n} q^{m-1}c_n(m) \tag{3}$$

This concludes the proof. □

Lemma 3.2.

$$-\Lambda_n'(q) = c_n(1) + c_n(2)q + c_n(3)q^2 + c_n(4)q^3 + \ldots = \sum_{m=1}^{\infty} c_n(m)q^{m-1}$$

Proof. This follows immediately from (3), using the fact that $c_n(k) = c_n(k+n)$, multiplying both sides by -1 and expanding out the geometric series to infinity. A quick illustration shows,

$$-\frac{1}{q^n-1}\sum_{m=1}^{n} q^{m-1}c_n(m) = \sum_{m=1}^{n}\left(\sum_{j=0}^{\infty}(q^n)^j\right)c_n(m)q^{m-1} = \sum_{m=1}^{\infty} c_n(m)q^{m-1}$$

□

Result 3.2.1. *The Von-Mangoldt function can be represented as the infinite series,*

$$-\Lambda(n) = c_n(1) + \frac{1}{2}c_n(2) + \frac{1}{3}c_n(3) + \frac{1}{4}c_n(4) + \dots$$

Proof. This follows from Lemma 3.2 by integration, we get

$$\int \Lambda_n'(q) = -\sum_{m=1}^{\infty}\frac{1}{m}c_n(m)q^m \Rightarrow -\Lambda_n(q) = c_n(1)q + \frac{1}{2}c_n(2)q^2 + \frac{1}{3}c_n(3)q^3 + \frac{1}{4}c_n(4)q^4 + \dots$$

then the result, which agrees with a very similar result discovered by Ramanujan in his 1918 paper, follows from $q \to 1$. This concludes the proof. □

q-Euler Totient Function

The Euler totient function defined as

$$\phi(n) = n\prod_{p|n}\left(1-\frac{1}{p}\right)$$

can be q-deformed from (1) by seeing that $\phi_n(1) = 2\Lambda_n'(1)$ and hence

$$\phi_n(q) = 2\Lambda_n'(q) \tag{4}$$

Result 3.2.2. *For $n \geq 2$,*

$$-\frac{1}{2}\phi(n) = c_n(1) + c_n(2) + c_n(3) + c_n(4) + \dots$$

Proof. This immediately follows from Lemma 3.2 and the fact that $\phi_n(1) = 2\Lambda_n'(1)$, concluding the proof. □

Note that this seems to be an identity we haven't been able to find in literature yet, however, to avoid being on the naïve side, it is unlikely a completely new discovery, but more likely a re-discovery.

Claim 3.2.1. *We claim, in order to demonstrate the power of this q-deformation that the famous*

$$\sum_{d|n} \phi(d) = n$$

$$\Rightarrow \sum_{\substack{d|n \\ d>1}} \phi(d) = n - 1$$

holds for $\phi_n(q)$. This seems obvious as otherwise there would be no point in this, however considering one could alternatively have defined a q-Euler totient function from 'scratch' instead of from (1), which would look slightly different, we demonstrate (as we have done by proving the identities) that staying consistent with (1) allows for subsequent discovery to be made with much greater foundation than from scratch. With this we mean that, for example, knowing

$$\phi(n) = n \sum_{d|n} \frac{\mu(d)}{d}$$

one could define from scratch, without deriving it from (1) that

$$\phi_n(q) = [n] \sum_{d|n} \frac{\mu(d)}{[d]}$$

and then in order to study it, inevitably use (1). This would introduce contradiction as an already naturally defined $\phi_n(q)$ awaits discovery as derivable from (1). This seems to suggest this q-deformation of number theory to be a very natural and very useful one.

Proof. Start with

$$\sum_{\substack{d|n \\ d>1}} \phi_d(q) = 2 \sum_{\substack{d|n \\ d>1}} \Lambda'_d(q) = \sum_{\substack{d|n \\ d>1}} \phi_d(q) = 2 \frac{d}{dq} \sum_{\substack{d|n \\ d>1}} \ln \Phi_d(q)$$

$$= 2 \frac{d}{dq} \ln \prod_{\substack{d|n \\ d>1}} \Phi_d(q)$$

We see that by (1) this is equal to

$$2 \frac{d}{dq} \ln [n] = 2(1 + 2q + 3q^2 + \ldots + (n-1)q^{n-2}) \frac{1}{[n]}$$

letting $q \to 1$ we see that the sum in brackets is the $(n-1)$-th triangle number which equals $n(n-1)/2$. Using this we can see that the above expression yields $n-1$, and so

$$\sum_{\substack{d|n \\ d>1}} \phi_d(1) = n-1$$

This concludes the proof. $\square$

Hopefully this demonstrated the 'natural-ness' of this q-deformation, and deemed it of non-trivial disposition unlike one created from 'scratch'. The derivative of $\phi_n(q)$ is also of interest, and can be calculated using a result from Theorem 1.1 in [2].

Claim 3.2.2. *We claim that for $n \geq 2$,*

$$\phi_n'(1) = \frac{J_2(n)}{6} - \phi(n)$$

where $J_2(n)$ is the 2nd Jordan totient function.

Proof. Recall,

$$\phi_n(q) = 2\Lambda_n'(q)$$

Taking the derivatives of both sides we can see that

$$\phi_n'(q) = 2\Lambda_n''(q) = 2\frac{d}{dq}\left(\frac{\Phi_n'(q)}{\Phi_n(q)}\right) = 2\frac{\Phi_n''(q)}{\Phi_n(q)} - 2\left(\frac{\Phi_n'(q)}{\Phi_n(q)}\right)^2$$

$$= 2g_n(q) - 2\Lambda_n'(q)^2$$

where g is some mystery function for now. By Theorem 1.1 [2] one sees that

$$g_n(1) = \frac{\phi(n)}{4}\left(\phi(n) + \frac{\Psi(n)}{3} - 2\right)$$

where $\Psi(n)$ is the Dedekind psi function, and therefore

$$\phi_n'(1) = 2g_n(1) - \frac{\phi_n(1)^2}{2} = \frac{\phi(n)}{2}\left(\phi(n) + \frac{\Psi(n)}{3} - 2\right) - \frac{\phi(n)^2}{2}$$

$$= \frac{\phi(n)^2}{2} + \frac{\phi(n)\Psi(n)}{6} - \phi(n) - \frac{\phi(n)^2}{2} = \frac{J_2(n)}{6} - \phi(n)$$

where we used the fact that $\phi(n)\Psi(n) = J_2(n)$. This concludes the proof. $\square$

Dividing both sides by $\phi(n)$ shows us the interesting form,

$$\frac{\phi_n'(1)}{\phi(n)} = \frac{\Psi(n)}{6} - 1 \tag{5}$$

suggesting the logarithmic derivative holds this equality, at least at $q = 1$.

Claim 3.2.3.

$$2g_n(q) = \phi_n(q)\frac{\Phi_n''(q)}{\Phi_n'(q)}$$

Proof. We can see if we take a look at the logarithmic derivative of $\phi_n(q)$ that we get the following,

$$\phi_n'(q) = \phi_n(q)\frac{\Phi_n''(q)}{\Phi_n'(q)} - \frac{\phi_n(q)^2}{2}$$

Comparing with

$$\phi_n'(q) = 2g_n(q) - 2\Lambda_n'(q)^2$$

we can see therefore that

$$2g_n(q) = \phi_n(q)\frac{\Phi_n''(q)}{\Phi_n'(q)}$$

This concludes the proof. $\square$

For amusement we can consider higher order derivatives,

$$\Lambda_n'''(q) =^? -\frac{\phi(n)^3}{4} - \frac{\phi(n)^2\Psi(n)}{4} + \frac{\phi(n)^2}{2} - \frac{J_2(n)}{4} + 1$$

q-Dedekind Psi Function

The Dedekind psi function has very interesting connections to number theory, and as the rest of the arithmetic functions arise out of (1). From (5) we can see that the q-Dedekind psi function $\Psi_n(q)$ is defined as,

$$\frac{\Psi_n(q)}{6} = \frac{\phi_n'(q)}{\phi_n(q)} + 1 \tag{6}$$

Result 3.2.3.

$$\frac{\Lambda_n'(q)\Psi_n(q)}{3} = c_n(1)+c_n(2)q+2c_n(3)q+c_n(3)q^2+3c_n(4)q^2+c_n(4)q^3+4c_n(5)q^3+\dots$$

Proof. The proof is left to the reader. $\square$

Result 3.2.4. *We obtain the rather beautiful result for the Jordan totient function $J_2(n)$,*

$$\frac{J_2(n)}{6} = c_n(1) + c_n(2) + 3c_n(3) + 4c_n(4) + 5c_n(5) + 6c_n(6) + 7c_n(7) + \dots$$

Proof. This follows from expanding the RHS in (6), using Lemma 3.2, and the fact that $\Psi_n(1)\phi_n(1) = J_2(n)$. $\square$

q-Chebyshev Function

We will focus in this section on the q-analog of the ψ Chebyshev function defined as

$$\psi(n) = \sum_{1 \le k \le n} \Lambda(k)$$

Naturally it would be

$$\psi_n(q) = \sum_{1 < k \le n} \Lambda_k(q) \tag{7}$$

The function $f_n(q) = e^{\psi_n(q)}$ has many of its own fascinating properties and number theoretic relations to other functions. We will go over a few briefly.

Lemma 3.3. *The coefficients of $f_n(q)$ are symmetric*

Proof. Start with the fact that $\Phi_n(x)$ has symmetric coefficients for $n > 1$. This is illustrated as

$$\Phi_n(x) = x^{\phi(n)} \Phi_n\left(\frac{1}{x}\right)$$

for $n > 1$. Hence,

$$f_n(q) = e^{\psi_n(q)} = \prod_{k=2}^{n} \Phi_k(q) = \prod_{k=2}^{n} q^{\phi(k)} \Phi_k\left(\frac{1}{q}\right)$$

Letting $D(n)$ denote the degree of $f_n(q)$, then

$$f_n(q) = \prod_{k=1}^{n} \Phi_k(q) = q^{D(n)} \prod_{k=1}^{n} \Phi_k\left(\frac{1}{q}\right) = q^{D(n)} f_n\left(\frac{1}{q}\right)$$

this concludes the proof. □

Many interesting facts about the Chebyshev function $\psi(n)$ can probably be derived from $f_n(q)$. From the above we can immediately see that the exponential of the Chebyshev function $\psi(n)$ is always even due to its symmetrical coefficients, a fact however, easily derived from the fact that it is a primorial therefore having a prime factor 2. Regardless of this, this is an adequate illustration of how many facts of arithemtic functions as well as maybe larger structures can be obtained from q-derivations, some of which might not be as easily derived as the former example.

Claim 3.3.1. *We will prove the claim that the coefficients of $f_n(q)$, which we will denote as $a_n(j)$, do not always form a unimodal sequence.*

Proof. By Vieta's formula and Lemma 3.3, one can see quickly that

$$\zeta_2 + \ldots + \zeta_3 + \zeta_3^2 + \ldots + \zeta_{D(n)} = a_n(D(n)-1) = a_n(1)$$

$$= \sum_{k=1}^{n} \mu(k) = M(n)$$

where $M(n)$ is the Mertens function. The first value for which $\mid M(n) \mid$ is lowest for $n > 2$, is for $M(39)$, and so we have then that $a_{39}(0), a_{39}(1), a_{39}(2), \ldots, a_{39}(D(n)))$ is not a unimodal sequence. This conludes the proof.

□

q-Totient Summatory Function

We can see that

$$\psi_n'(q) = \sum_{1<k\leq n} \Lambda_k'(q) = \frac{1}{2} \sum_{1<k\leq n} \phi_k(q)$$

so therefore

$$\psi_n'(1) = \frac{T(n)-1}{2}$$

where $T(n) = \sum_{1\leq k\leq n} \phi(k)$ is the totient summatory function, and therefore

$$T_n(q) = 2\psi_n'(q) + 1 = \frac{2f_n'(q)}{f_n(q)} + 1 \tag{8}$$

4 References

[1] Ramanujan, On Certain Trigonometric Sums & Their Applications in Number Theory

[2] Coefficients & Higher Order Derivatives of Cyclotomic Polynomials